

Influence of pacing strategy on O₂ uptake and exercise tolerance

A. M. Jones¹, D. P. Wilkerson¹, A. Vanhatalo², M. Burnley²

¹School of Sport and Health Sciences, University of Exeter, Exeter, Devon, UK, ²Department of Sport and Exercise Science, University of Wales Aberystwyth, Aberystwyth, Ceredigion, UK

Corresponding author: Andrew M. Jones, PhD, School of Sport and Health Sciences, St. Luke's Campus, University of Exeter, Heavitree Road, Exeter, Devon, EX1 2LU, UK. Tel: 01392 262886, Fax: 01392 264726, E-mail: a.m.jones@exeter.ac.uk

Accepted for publication 28 June 2007

Seven male subjects completed cycle exercise bouts to the limit of tolerance on three occasions: (1) at a constant work rate (340 ± 57 W; even-pace strategy; ES); (2) at a work rate that was initially 10% lower than that in the ES trial but which then increased with time such that it was 10% above that in the ES trial after 120 s of exercise (slow-start strategy; SS); and, (3) at a work rate that was initially 10% higher than that in the ES trial but which then decreased with time such that it was 10% below that in the ES trial after 120 s of exercise (fast-start strategy; FS). The expected time to exhaustion predicted from the pre-established power–time relationship was 120 s in all three

conditions. However, the time to exhaustion was significantly greater ($P < 0.05$) for the FS (174 ± 56 s) compared with the ES (128 ± 21 s) and SS (128 ± 30 s) conditions. In the FS condition, $\dot{V}O_2$ increased more rapidly toward its peak such that the total O₂ consumed in the first 120 s of exercise was greater (ES: 5.15 ± 0.78 ; SS: 5.07 ± 0.83 ; FS: 5.36 ± 0.84 L; $P < 0.05$ for FS vs ES and SS). These results suggest that a fast-start pacing strategy might enhance exercise tolerance by increasing the oxidative contribution to energy turnover and hence “sparing” some of the finite anaerobic capacity across the transition to high-intensity exercise.

The manner in which athletes distribute their energy expenditure during a race, i.e. the pacing strategy they choose, is likely to influence factors such as the contribution to energy turnover from oxidative and non-oxidative metabolic pathways, the rate at which fatigue develops and ultimately the race performance itself. It is surprising, therefore, that comparatively few studies have specifically investigated the influence of pacing strategy on exercise performance (Foster et al., 1994; Fukuba & Whipp, 1999).

Despite the relative dearth of experimental evidence, the prevailing view has been that an even-paced strategy is optimal for athletic performance (Robinson et al., 1958; Leger & Ferguson, 1974; Ariyoshi et al., 1979b; Foster et al., 1993; Thompson et al., 2003 see Foster et al., 1994, for a review). However, recent studies indicate that, for high-intensity exercise of approximately 1–5 min duration, a “fast-start” or “all-out” pacing strategy results in a better performance outcome than an even-pace strategy (de Koning et al., 1999; Bishop et al., 2002). For example, Bishop et al. (2002) reported that an all-out pacing strategy resulted in a greater average power output during a 2-min kayak ergometer test, with no change in the measured peak $\dot{V}O_2$, peak blood [lactate], or accumulated O₂ deficit (AOD). The explanation for this effect is not entirely clear, but

has been suggested to be related to the effectiveness with which the available anaerobic energy supply is distributed during the race (Bishop et al., 2002; Foster et al., 2003; Hettinga et al., 2006).

During constant-load exercise above the asymptote of the hyperbolic power–time (P – t) relationship [the so-called critical power (CP)], the termination of exercise is believed to coincide with the depletion of the finite “anaerobic” work capacity (W'), notionally equivalent to an energy store consisting of O₂ reserves (myoglobin, venous blood), high-energy phosphates, and a source related to anaerobic glycolysis (Monod & Scherrer, 1965; Moritani et al., 1981; Poole et al., 1988; Miura et al., 1999; Coats et al., 2003; Fukuba et al., 2003). The higher the work rate above the CP, the more rapidly W' will be expended and the greater will be the rate at which metabolites that have been associated with the fatigue process (e.g. P_i , H^+ , extra-cellular K^+ ; Westerblad & Allen, 2003; Robergs et al., 2004) accumulate. However, athletes might choose to expend the available “anaerobic” resource in a number of ways according to the competitive situation (i.e., with a fast start, a fast finish or with a high constant pace throughout the event; Fukuba & Whipp, 1999). Strategies that maximize the contribution of oxidative metabolism to energy turnover during a bout of high-intensity

exercise, while still allowing the W' to be exhausted, should allow more work to be done and thus result in superior exercise performance.

The absolute rate at which oxygen uptake ($\dot{V}O_2$) rises following the onset of exercise is a positive function of the “error signal,” i.e. the difference between the current and the “required” steady-state $\dot{V}O_2$ (Margaria et al., 1965; Whipp & Mahler, 1980). In other words, the increase in $\dot{V}O_2$ following the onset of exercise is driven by the instantaneous rate of ATP hydrolysis in the contracting myocytes, which is itself reflected in the rate of change of intramuscular phosphocreatine concentration ([PCr]) per unit change in time (t), i.e. $\Delta[\text{PCr}]/\Delta t$ (Rossiter et al., 2005). A “fast-start” pacing strategy, which would magnify the $\dot{V}O_2$ error signal, might therefore be hypothesized to result in a more rapid absolute increase in $\dot{V}O_2$ following the onset of exercise and possibly to enhance performance, with the proviso that the W' is not prematurely depleted. To date, studies examining the physiological responses to, and performance effects of, different pacing strategies have not focused specifically on possible changes in the dynamic adaptation of $\dot{V}O_2$ during exercise (Foster et al., 1993; Bishop et al., 2002; Hettinga et al., 2006).

The limited available evidence suggests that changes in the temporal distribution of the work done, where the average work rate is above CP and exercise is continued to exhaustion, do not alter exercise performance appreciably (Fukuba et al., 2003). These data indicate that the available W' and the average rate at which it declines are similar during constant work rate and variable work rate exercise performed above the CP (Fukuba et al., 2000). However, such conclusions might be considered to be incompatible with recent reports that pacing strategy significantly alters the tolerance to high-intensity exercise (de Koning et al., 1999; Bishop et al., 2002). Indeed, these latter results potentially challenge one of the essential tenets of the CP concept, namely that the W' represents a finite amount of work that can be done above the CP that is fixed from the onset of exercise.

The purpose of the present study was therefore to examine the influence of pacing strategy (even paced, ES; slow start, SS; and fast start, FS) on $\dot{V}O_2$ response dynamics and performance during supra-maximal cycle exercise where the expected tolerable duration of exercise, as predicted from the pre-established P - t relationship, was 120 s (a similar duration to the 800 m track race). We hypothesized that FS would result in a faster absolute rise in $\dot{V}O_2$ (as assessed by the “mean response time,” MRT) and an augmented oxidative contribution to energy turnover (see Fig. 1 for schematic illustration of the rationale for this investigation). With the assumption

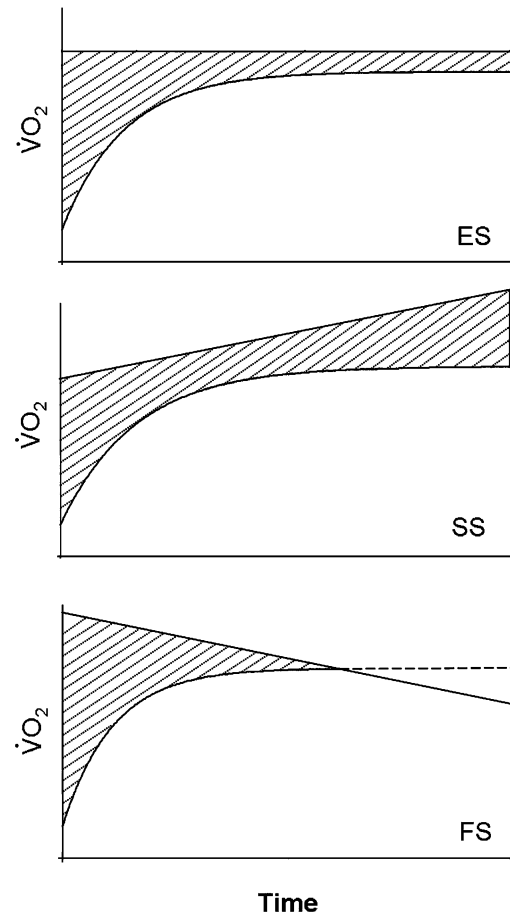


Fig. 1. Schematic illustration of the rationale for the investigation. Three different pacing strategies (even-pace, ES; slow-start, SS; and fast-start, FS) are represented in the upper, middle, and lower panels, respectively. The average work rate and thus the total work done in the first 120 s of exercise is assumed to be the same for all three conditions. In all figures, the straight line at the top represents the “steady-state” $\dot{V}O_2$ that would be required for the exercise task (i.e., the energy demand expressed in units of $\dot{V}O_2$) and the curved line represents the actual $\dot{V}O_2$ response to the exercise. Note that because the exercise task is deliberately “supra-maximal” (requiring a time to exhaustion of approximately 120 s), $\dot{V}O_2$ can never reach the demand but will be truncated at the $\dot{V}O_2$ max. The difference between the straight line (energy demand) and the curved line (energy supply through increased O_2 uptake) is the O_2 deficit (shaded area). Relative to ES, the absolute $\dot{V}O_2$ response is hypothesized to be slower in SS and faster in FS (due to the different work rates and thus muscle ATP turnover rates at the start of exercise in the three conditions). If true, this would lead to differences in the magnitude of the O_2 deficit incurred (greatest in SS and least in FS) and – importantly – the extent to which the anaerobic work capacity (W') is expended, and possibly have important implications for exercise tolerance. In FS, the dashed line suggests that beyond a certain time, the actual $\dot{V}O_2$ might be higher than the $\dot{V}O_2$ demand if $\dot{V}O_2$ max is maintained until the end of exercise.

that the average ATP turnover rate was constant between the three conditions, we further hypothesized that the increased oxidative contribution to

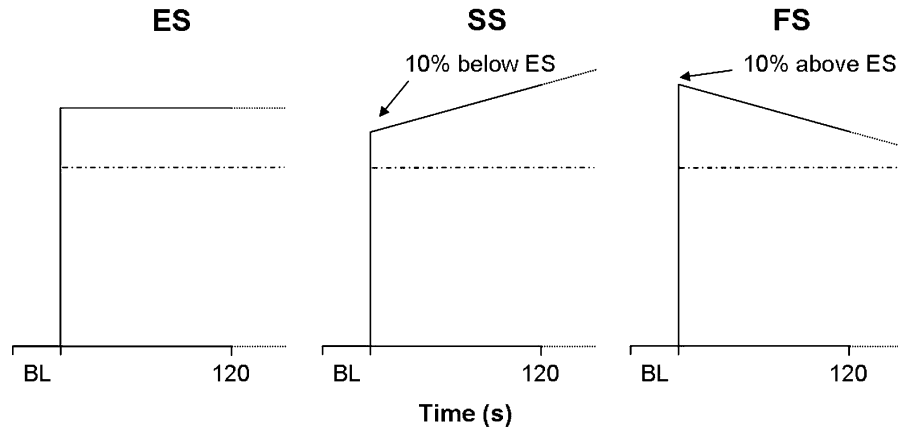


Fig. 2. Schematic illustration of the experimental protocol. All conditions began with an abrupt increase to the initial target work rate from a baseline of cycling at 20 W. In the even-pace (ES) condition, the target work rate (estimated to result in exhaustion in 120 s) was held constant throughout exercise. In the slow-start (SS) condition, the work rate was initially increased to a value 10% lower than that in ES and then increased progressively with time until it was 10% above that in ES after 120 s of exercise. In the fast-start (FS) condition, the work rate was initially increased to a value 10% higher than that in ES and then decreased progressively with time until it was 10% below that in ES after 120 s of exercise. In this way, the total work done was the same for the first 120 s of exercise in all conditions. The dotted line indicates that the protocols continued to the individual limit of tolerance. The horizontal broken line represents the CP.

energy turnover in FS would reduce the rate at which W' was depleted and thus result in improved exercise tolerance (measured as an extended time to exhaustion; i.e. beyond 120 s).

Methods

Subjects

The subjects were seven healthy males (mean $\pm$ SD; age, 22 ± 3 years; stature, 1.79 ± 0.08 m; body mass, 75.3 ± 9.5 kg) who were recreationally active in sporting activities but were not highly trained. All subjects gave written informed consent to participate in the study, which had received approval from the institutional research ethics committees, following an explanation of the benefits and risks of participation. The subjects visited the exercise physiology laboratory on seven occasions within a period of 21 days. Exercise tests were separated by a minimum of 48 h. On each occasion, the subjects were required to report to the laboratory well hydrated, at least 2 h following the consumption of a light meal, and having completed no strenuous exercise within the preceding 24 h.

Procedures

All tests were conducted on an electrically braked cycle ergometer (Excalibur Sport, Lode, the Netherlands) in an air-conditioned laboratory with the temperature set at 18 °C. On their first visit to the laboratory, the subjects were familiarized with the experimental procedures and apparatus and they completed a ramp incremental test to exhaustion for the determination of $\dot{V}O_2$ peak (defined as the mean $\dot{V}O_2$ over the last 30 s of exercise), gas exchange threshold (determined using the V-slope method) and delta efficiency (i.e., $\Delta\dot{V}O_2/\Delta W/R$ slope) (Whipp et al., 1981; Wilkerson et al., 2004a). This test commenced with 3 min of baseline cycling at 20 W before the work rate was increased at a rate of 30 W/min until the limit of tolerance. In this initial test, each subject selected a pedal rate between 70 and 90 rev./min; this same pedal rate (± 2 rev./min) was maintained in all subsequent tests. The

height of the seat and handlebar were adjusted for comfort for each subject and recorded and reproduced in all subsequent tests.

During visits 2–4, the subjects completed, in random order, step exercise tests to volitional exhaustion at 85%, 90%, and 95% of the peak work rate achieved in the ramp incremental test. These intensities were chosen to elicit times to fatigue ranging from 2 to 10 min. The purpose of these tests was to determine the parameters of the hyperbolic P - t relationship (i.e., CP and W' ; Moritani et al., 1981; Poole et al., 1988). These tests also began with 3 min of pedaling at 20 W before the work rate was abruptly increased to the target work rate. A hand-held stopwatch was used to record the time to exhaustion to the nearest second. The time to exhaustion was defined as the time at which the pedal rate fell more than 5 rev./min below the individual's self-selected pedal rate.

The CP and W' were calculated according to the linear model of work rate vs $1/\text{time}$ in which the slope of the relationship provides the curvature constant, W' , and the y -axis intercept corresponds to CP (Poole et al., 1988; Fukuba et al., 2003). The work rate, which would be expected to lead to exhaustion in 120 s, was then calculated from the equation:

$$P = (W'/T_E) + \text{CP} \quad [1]$$

where P is the target work rate, W' is the anaerobic work capacity in Joules (J), T_E is the time to exhaustion (set at 120 s), and CP is the critical power. For example, the work rate estimated to result in a time to exhaustion of 120 s in a subject with W' of 15 000 J and CP of 225 W would be $(15\,000/120) + 225 = 350$ W.

During visits five to seven, the subjects completed, in random order, three "step" exercise tests to the limit of tolerance in which the work rate was either held constant throughout (i.e., even-paced strategy, ES) or controlled to result in either a slow start (SS) or fast start (FS) pacing strategy, but with the total work done held constant for the first 120 s of each condition (Fig. 2). With this experimental design, the W' should be exhausted in 120 s and should lead to the termination of exercise at 120 s in all three conditions, according to the CP concept (Monod & Scherrer, 1965; Moritani et al., 1981; Poole et al., 1988; Miura et al., 1999; Coats et al., 2003; Fukuba et al., 2003). All conditions

commenced with 3 min of baseline pedaling at 20 W before an abrupt transition to the initial target work rate, which was achieved within ~ 3 s. In ES, the target work rate was held constant at that calculated to result in exhaustion in 120 s (e.g. 350 W in the example given above). In SS, the work rate was abruptly increased to a work rate that was initially 10% below the target work rate (i.e., 315 W in the example) and then was gradually increased with time such that it was 10% above the target work (i.e., 385 W in the example) at 120 s of exercise (i.e., the work rate increased by 0.58 W every second). In FS, the initial work rate was 10% higher than the target work rate and it decreased with time such that it was 10% lower than the target work rate at 120 s of exercise. For all protocols, the cycle ergometer was programmed to control the work rate: a step increase to the initial work rate and then remaining constant, increasing with time or decreasing with time for ES, SS and FS, respectively.

It should be noted that the total work done in the first 120 s of exercise was identical in all three conditions. Also, because the average work rate was considerably above the CP, the time to exhaustion would be expected to coincide with the depletion of W' in all conditions (and to occur at ~ 120 s, unless pacing strategy influenced either the average energy turnover rate or the oxidative contribution to ATP production in this period). However, in each case, the protocols were programmed to continue for 5 min, and the subjects received strong verbal encouragement to continue to exhaustion. The subjects were naïve to the purpose of the study and were blinded to the time elapsed during exercise. A hand-held stopwatch was used to record the time to exhaustion to the nearest second. Time to exhaustion was defined as the point at which the pedal rate fell more than 5 rev./min below the subject's self-selected pedal rate. No "warm-up" was performed before trials 2–7 because it has been shown that prior exercise can profoundly alter the $\dot{V}O_2$ response dynamics and exercise performance (Jones et al., 2003b; Burnley et al., 2005).

Pulmonary gas exchange and ventilation were determined on a breath-by-breath basis throughout all of the exercise tests, except for those used for the determination of CP, with a metabolic cart (Jaeger Oxycon Pro, Hoechberg, Germany). Before each test, the system was calibrated using gases of known concentration and a 3 liters calibration syringe over a range of flows. The time delay between the volume and gas concentration signals was measured and used to phase-align the signals. Heart rate (HR) was recorded every 5 s throughout the exercise tests using short-range telemetry (Polar PE 4000, Polar, Finland). During the three experimental conditions (i.e., ES, SS and FS), a blood sample from a fingertip was collected into a capillary tube over the last 20 s of the period of baseline cycling, within 20 s of the termination of exercise and after 5 min of recovery, and analyzed for whole blood [lactate] using an automated analyzer (YSI 1500 Sport, Yellow Springs Instruments, Ohio, USA). Δ blood [lactate] was calculated as the difference between the pre- and post-test blood [lactate].

To provide an indication of the response dynamics of $\dot{V}O_2$ following the onset of exercise with each of the three pacing strategies, we calculated the MRT for $\dot{V}O_2$ by fitting a mono-exponential curve to the data from the onset of exercise (Whipp & Rossiter, 2005):

$$\dot{V}O_2(t) = \dot{V}O_2(b) + A \cdot (1 - e^{-t/\tau}) \quad [2]$$

where $\dot{V}O_2(t)$ is $\dot{V}O_2$ at any time t , $\dot{V}O_2(b)$ is the average $\dot{V}O_2$ in the last 2 min of the baseline period, A is the asymptote of the increase in $\dot{V}O_2$ above baseline (functionally the $\dot{V}O_{2max}$) and τ is the time constant (equivalent to the MRT in this model). With only one transition performed in each condition, more complex models were not considered suitable. And, because all three protocols resulted in the rapid attainment

of the $\dot{V}O_2$ peak, a single exponential function commencing at the onset of exercise was considered to be the most appropriate approach for characterizing the overall $\dot{V}O_2$ kinetics (Wilkerson et al., 2004a; Whipp & Rossiter, 2005). In addition, we calculated the total O_2 consumed for: (1) the first 120 s of exercise; and (2) the entire duration of exercise, in each condition. Finally, we calculated the AOD as the difference between the total O_2 consumed and the total O_2 required during the entire exercise period, with the latter estimated from individual exercise economy characteristics (derived from the $\Delta\dot{V}O_2/\Delta WR$ slope in the linear portion of the ramp incremental test) and the time to exhaustion.

Statistics

Differences in physiological response and in time to exhaustion between the three pacing strategies were assessed using analysis of variance with *post hoc* Bonferroni-adjusted *t*-tests as appropriate. Statistical significance was accepted when $P < 0.05$. The Pearson product moment correlation coefficient was used to examine the relationship between changes in $\dot{V}O_2$ dynamics and the AOD and changes in exercise performance. Data are presented as mean $\pm$ SD unless otherwise indicated.

Results

Ramp incremental test and power-duration relationship

During the ramp incremental test, the subjects achieved a $\dot{V}O_2$ peak of 50.6 ± 8.1 mL/kg/min and attained a peak work rate of 330 ± 40 W. The GET occurred at $55 \pm 4\%$ of the $\dot{V}O_2$ peak. The work rates corresponding to 85%, 90%, and 95% of the peak work rate attained in the ramp incremental test were 280 ± 34 , 297 ± 37 and 314 ± 39 W, respectively. The correlation coefficients between work rate and time to exhaustion were high for all subjects (range: 0.91–0.99). From these trials, the CP and W' were calculated to be 238 ± 29 W and 12.8 ± 3.2 kJ, respectively. Using eqn. [1] (see Methods), the constant work rate that would result in exhaustion in 120 s was estimated to be 344 ± 52 W (i.e., $\sim 104\%$ of the peak work rate attained in the ramp test, and $\sim 145\%$ of the CP).

Effect of pacing strategies on physiological responses and exercise tolerance

The principal results are presented in Table 1 and illustrated for a representative subject in Fig. 3. The time to exhaustion for ES (128 ± 28 s) and SS (128 ± 19 s) were not significantly different from one another and were close to (and not significantly different from) the predicted time to exhaustion of 120 s. However, the time to exhaustion in FS (174 ± 52 s) was significantly greater than 120 s ($P < 0.05$), and also significantly greater than ES and SS ($P < 0.05$). The total work done before exhaustion was, on average, 44.0 kJ for ES, 44.3 kJ for SS, and 57.2 kJ for FS. The total work done above CP before exhaustion was not significantly

Table 1. Mean $\pm$ SD physiological responses to high-intensity exercise performed with different pacing strategies

	Even-pace	Slow-start	Fast-start
Time to exhaustion (s)	128 $\pm$ 28	128 $\pm$ 19	174 $\pm$ 52*
Baseline $\dot{V}O_2$ (L/min)	0.83 $\pm$ 0.12	0.85 $\pm$ 0.10	0.84 $\pm$ 0.10
Peak $\dot{V}O_2$ (L/min)	3.68 $\pm$ 0.43	3.66 $\pm$ 0.35	3.69 $\pm$ 0.35
Peak HR (b/min)	184 $\pm$ 7	183 $\pm$ 5	187 $\pm$ 8
MRT-120 (s)	50 $\pm$ 10	54 $\pm$ 10	37 $\pm$ 7*
MRT-overall (s)	50 $\pm$ 11	53 $\pm$ 11	40 $\pm$ 13 [†]
O ₂ consumed-120 (L)	5.15 $\pm$ 0.78	5.07 $\pm$ 0.83	5.36 $\pm$ 0.84*
AOD-120 (L)	2.49 $\pm$ 0.82	2.54 $\pm$ 0.46	2.34 $\pm$ 0.80*
O ₂ consumed-overall (L)	6.17 $\pm$ 1.66	6.06 $\pm$ 0.70	9.15 $\pm$ 3.00*
AOD-overall (L)	1.98 $\pm$ 0.70	2.10 $\pm$ 0.49	0.90 $\pm$ 1.63*
Work done above CP (kJ)	13.4 $\pm$ 2.9	14.0 $\pm$ 2.3	15.7 $\pm$ 3.1*
Δ Blood [lactate] (mM)	7.3 $\pm$ 1.5	6.9 $\pm$ 2.0	6.8 $\pm$ 1.2

*Significantly different from the even-pace and slow-start conditions ($P < 0.05$).

[†]Significantly different from slow-start condition ($P < 0.05$).

MRT-120, mean response time of $\dot{V}O_2$ when data were fit to 120 s or only to the time reached in the shortest trial; MRT-overall, mean response time of $\dot{V}O_2$ when the entire response was modelled; O₂ consumed-120, total volume of O₂ consumed in the first 120 s or the time reached in the shortest trial; O₂ consumed-overall, total volume of O₂ consumed throughout exercise; AOD-120, accumulated O₂ deficit in the first 120 s or the time reached in the shortest trial; AOD-Overall, accumulated O₂ deficit throughout exercise; HR, heart rate; CP, critical power.

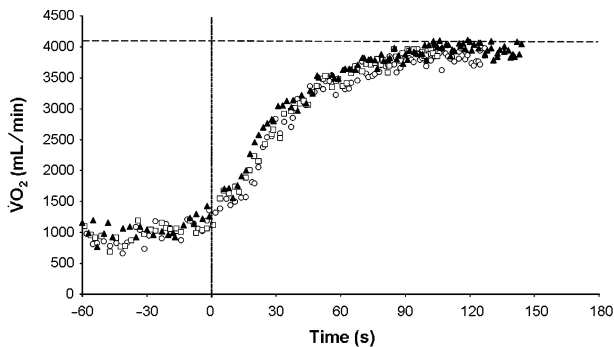


Fig. 3. Pulmonary O₂ uptake response with each of the three different pacing strategies in a representative subject. Open circles represent slow-start, open squares represent even-pace, and closed triangles represent fast-start (FS). The step increase in work rate occurred at time zero. The dashed horizontal line represents the subject's $\dot{V}O_2$ peak as measured during the initial ramp incremental exercise test. Note the increased $\dot{V}O_2$ over the first 60–90 s of exercise and the extended time to exhaustion in the FS condition.

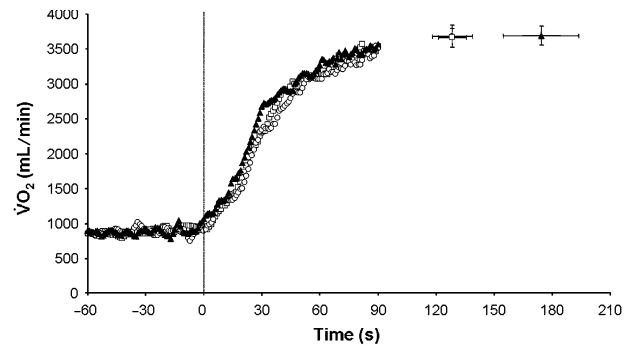


Fig. 4. Group mean $\dot{V}O_2$ response with each of the three pacing strategies for the first 90 s of exercise (the shortest time to exhaustion recorded). Open circles represent slow-start (SS), open squares represent even-pace (ES), and closed triangles represent fast-start. The step increase in work rate occurred at time zero. The mean $\pm$ SEM $\dot{V}O_2$ peak and time to exhaustion are also shown (the points for SS and ES overlaid).

different between ES and SS but was significantly higher for FS (Table 1). The peak $\dot{V}O_2$ attained at the end of exercise was not significantly different between the three conditions (ES: 3.68 $\pm$ 0.43 L/min; SS: 3.66 $\pm$ 0.35 L/min; FS: 3.69 $\pm$ 0.35 L/min) and was similar to the $\dot{V}O_2$ peak measured during the ramp incremental exercise test (3.80 $\pm$ 0.42; $P > 0.05$).

The MRT was significantly shorter during FS compared with ES and SS when the data were modeled to 120 s or only to the time reached in the shortest trial, and significantly shorter during FS compared with SS when the entire response was considered. The faster adjustment of $\dot{V}O_2$ in the FS condition resulted in there being a significantly greater consumption of O₂ over the first 120 s of

exercise in this condition (ES: 5.15 $\pm$ 0.78 liters; SS: 5.07 $\pm$ 0.83 liters; FS: 5.35 $\pm$ 0.84 liters; $P < 0.05$ for FS vs SS and ES) and therefore a reduced O₂ deficit. The mean $\dot{V}O_2$ response for the group with each of the three pacing strategies is shown in Fig. 4. The percentage reduction in the MRT was significantly correlated with the percentage improvement in time to exhaustion when FS was compared with ES ($r = 0.85$, $P < 0.05$; Fig. 5). The energy equivalent of the additional O₂ consumed over the first 120 s of exercise in the FS condition compared with the ES condition (i.e. the difference in the total O₂ consumed in first 120 s of exercise between FS (5.36 L) and ES (5.15 L) multiplied by 21 kJ and assuming 25% efficiency = 5.36 – 5.15 $\cdot$ 21 kJ $\cdot$ 0.25) was 1.1 $\pm$ 1.0 kJ.

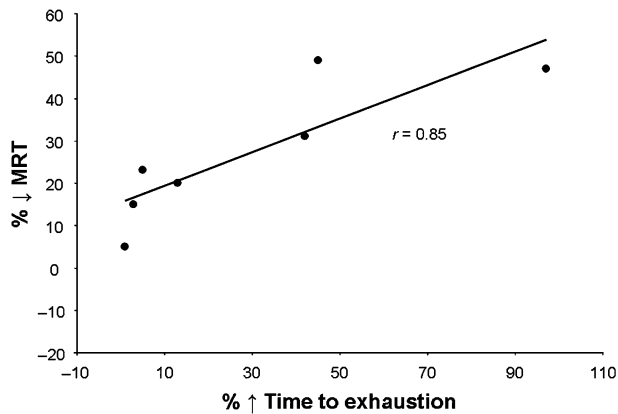


Fig. 5. Relationship between the percentage reduction in the mean response time (MRT) in fast-start (FS) compared with even-pace (ES) and the percentage increase in the time to exhaustion in FS compared with ES. The correlation coefficient of $r = 0.85$ was statistically significant ($P < 0.05$).

This was not significantly different from the additional work done above CP in FS compared with ES (2.4 ± 1.8 kJ), but the two variables were not significantly correlated ($r = 0.55$, $P = 0.21$).

The blood [lactate] values were not significantly different between the three pacing conditions at baseline, immediately post-exercise, or following 5 min of recovery (Table 1). Interestingly, however, the AOD at the end of exercise was similar between ES and SS but was significantly smaller ($P < 0.05$) during FS (ES: 1.35 ± 0.61 ; SS: 1.45 ± 0.40 ; FS: 0.23 ± 1.61 L). During FS, the total AOD was negatively correlated with the time to exhaustion ($r = -0.97$, $P < 0.001$; Fig. 6).

Discussion

The results of the present study show that a fast-start pacing strategy enhances the tolerance to high-intensity exercise by increasing the oxidative contribution to energy turnover early in exercise and enabling a greater amount of work to be done above the CP before exhaustion. It appears that invoking a larger “error signal” at the onset of exercise provides a greater stimulus to mitochondrial respiration such that $\dot{V}O_2$ rises more rapidly (in absolute terms) toward the anticipated $\dot{V}O_2$ requirement. Our results also appear to challenge one of the tenets of the CP concept, namely that the W' represents an immutable quantity of “anaerobic” work that can be performed above the CP.

Our study was designed so that the total work done was identical over the first 120 s of exercise. The average work rate used in all three pacing conditions was calculated from the pre-determined P - t relation-

ship to result in exhaustion in 120 s. However, while the time to exhaustion in the ES and SS conditions was not significantly different from 120 s, the time to exhaustion was considerably longer in the FS condition (i.e., 174 s, or 45% greater than the prediction). With the assumptions that (1) the total energy requirement was indeed identical in the three conditions for the first 120 s of exercise; and (2) exhaustion during exercise performed above CP coincides with the depletion of the W' (Monod & Scherrer, 1965; Moritani et al., 1981; Poole et al., 1988; Coats et al., 2003; Fukuba et al., 2003), this suggests that the increased consumption of O_2 over the first 120 s of exercise with the FS strategy reduced the rate at which the finite W' was depleted (and/or fatigue-related metabolites accumulated), allowing exercise to be prolonged. This is supported by the similarity of the energy equivalent of the increased O_2 consumption in the first 120 s of exercise and the additional work done above CP in FS compared with ES, and also by the significant correlation between the percentage reduction in MRT and the % increase in time to exhaustion in FS compared with ES (Fig. 5).

There was no significant difference in time to exhaustion between the ES and SS conditions. Given the significant extension of time to exhaustion in FS relative to ES, one might have expected a significantly shorter time to exhaustion in SS relative to ES. The MRT was longest and the total O_2 consumed in the first 120 s of exercise was least in the SS condition, but these differences did not achieve statistical significance. One explanation for these results might simply be the greater variability in performance during time to exhaustion trials compared with time trials in which subjects complete a given distance or amount of work in the least possible time (Hopkins et al., 1999). However, the coefficient of variation in time to exhaustion during supra-maximal exercise is $\sim 5\%$ in our laboratory (Wilkerson DP and Jones AM, unpublished observations), i.e. much less than the $\sim 36\%$ increase in time to exhaustion in FS compared with ES. Time to exhaustion trials were considered necessary in the present study because they afforded the opportunity to assess the volume of O_2 consumed for the same total work done when different pacing strategies were utilized.

It is, of course, also possible that there is a “non-linearity” in the influence of pacing strategy on $\dot{V}O_2$ dynamics and performance during supra-maximal exercise, i.e. that relative to an even-pace strategy, a fast-start strategy enhances $\dot{V}O_2$ and performance whereas a slow-start strategy does not significantly impair $\dot{V}O_2$ or performance. System non-linearities at intensities with a $\dot{V}O_2$ demand close to or above the $\dot{V}O_{2max}$ have been reported previously (Hughson et al., 2000; Özyener et al., 2001; Scheuermann &

Barstow, 2003; Wilkerson et al., 2004a,b; Wilkerson & Jones, 2006). Interestingly, asymmetrical gas exchange responses to incremental and decremental ramp exercise have been identified, albeit for moderate work rates (Miyamoto & Niizeki, 1992; Niizeki et al., 1995). These authors reported that the MRT for $\dot{V}O_2$ was significantly shorter for decremental exercise (similar to FS in the present study) than for incremental exercise (similar to SS in the present study) and suggested that this might have been a function of faster cardiac output kinetics or enhanced distribution of blood flow to the working muscles (Niizeki et al., 1995). It is possible that these effects contributed to the different $\dot{V}O_2$ and performance profiles observed in the present study.

Influence of pacing strategy on $\dot{V}O_2$ and blood [lactate]

In the present study, the highest $\dot{V}O_2$ attained was not significantly different between the pacing strategies and was very similar to (and not significantly different from) the $\dot{V}O_2$ peak measured during the ramp incremental test. The finding of a similar peak $\dot{V}O_2$ being attained with different pacing strategies is consistent with several previous studies (Ariyoshi et al., 1979a; Foster et al., 1993, 2004; Bishop et al., 2002; Aisbett et al., 2003; Fukuba et al., 2003). Thus, the greater total O_2 consumed in FS resulted not from any change in the highest $\dot{V}O_2$ attained but rather from enhanced $\dot{V}O_2$ dynamics. The absolute rate of change of $\dot{V}O_2$ following the onset of exercise is proportional to the imposed work rate and thus muscle ATP turnover rate (Margaria et al., 1965; Whipp & Mahler, 1980; Özyener et al., 2001; Wilkerson et al., 2004a). The latter is itself proportional to $\Delta[PCr]/\Delta t$ and therefore a greater rate of change of [PCr] at and following the onset of exercise should be associated with a more rapid increase in $\dot{V}O_2$ (Mahler, 1985; Rossiter et al., 2005). Muscle PCr acts as a buffer to maintain [ATP] following the onset of muscle contraction but its splitting results in the transduction, via the "PCr shuttle," of ADP to the mitochondrion where it serves as both a signal and a substrate for an acceleration of oxidative phosphorylation (Meyer et al., 1984; Mahler, 1985). Therefore, the FS pacing strategy used in the present study presumably increased the $\Delta[PCr]/\Delta t$ and provided a greater stimulus for muscle $\dot{V}O_2$, leading to faster overall $\dot{V}O_2$ dynamics and a greater total O_2 consumption during the first 120 s of exercise.

Our results are similar in certain respects to those of Ariyoshi et al. (1979a). Using Douglas bag collections of expired air, these authors reported that, relative to even-paced and slow-start strategies, a fast-start pacing strategy tended to increase $\dot{V}O_2$ at discrete time points across the transition from rest to exercise, and to reduce post-exercise blood [lactate]

and ratings of perceived exertion (Ariyoshi et al., 1979a); however, despite these potentially beneficial effects, a fast-start pacing strategy did not produce a statistically significant improvement in performance relative to an even-pace strategy (Ariyoshi et al., 1979b). Our results are consistent with the more recent study of Bishop et al. (2002) who reported that an "all-out" starting strategy enabled both a higher average power output and a higher total O_2 consumption (with no difference in peak $\dot{V}O_2$, AOD, or blood lactate accumulation) during a 2-min performance trial in well-trained kayakers. The results of Foster et al. (1994) also suggested that a fast-start strategy was associated with improved performance over 1500 m in elite speed skaters. Non-significant trends for a fast-start strategy to improve performance have also been noted by Aisbett et al. (2003) for 6 min of high-intensity cycle exercise and by Hettinga et al. (2006) for a 4000 m cycle time trial. Unfortunately, $\dot{V}O_2$ dynamics were not quantified in these studies. In contrast, Foster et al. (1993) found that an even-pace strategy produced better results than either a slow-start or fast-start strategy during a 2 km cycle time trial and Thompson et al. (2003) reported that even pacing was less physically stressful than fast-start or slow-start strategies during high-intensity swimming. The explanation for these differences between studies is not clear but might be related to subtle differences in exercise protocol and imposed pacing strategy.

Our results do not cohere with those of Fukuba et al. (2003). In that study, subjects performed bouts of exhaustive exercise during which the work rate was either switched from 117% to 134% CP ("UP"), or from 134% to 117% CP ("DOWN"), at the point at which approximately half of the W' should have been expended. The time to exhaustion was not reported but the total O_2 consumed during exercise appeared to be higher in the DOWN protocol (Fig. 4 in Fukuba et al., 2003); also, the actual W' was ~5% higher in the DOWN protocol (12.7 vs 12.1 kJ) but this was not statistically different. Fukuba et al. (2003) concluded that the work equivalent of W' (and presumably its rate of expenditure) is not affected by work rate variations at least in the range of intensities they studied. The most obvious difference between our study and that of Fukuba et al. (2003) is the exercise intensity range considered (130–159% CP in our study; 117–134% CP in theirs). It appears therefore that the differences in O_2 consumption and/or W' and its utilization, which are evident with a fast-start strategy, have greater potential to influence exercise tolerance during near-maximal exercise where the time to fatigue is ~2–3 min.

It is important to emphasize that, although the average work rate was only matched for the first 120 s of exercise in each of the conditions in the

present study, this alone cannot explain the increased work done above CP before exhaustion in the FS condition. This is because, according to the CP concept, the amount of work that can be done above the CP is finite (Monod & Scherrer, 1965); once exhausted, exercise can theoretically only be continued if some of the W' is reconstituted through rest or a period of lighter exercise performed <CP (Coats et al., 2003). Although the work rate continued to fall with time beyond 120 s in the FS condition, it remained substantially above the CP (i.e. ~30% above CP at 120 s, ~16% above CP at 180 s). Therefore, the ability of the subjects to continue to exercise beyond 120 s in the FS condition has to be explained by a pacing strategy-related enhancement of the W' and/or CP.

Consistent with previous studies (Foster et al., 1993; Bishop et al., 2002; Aisbett et al., 2003; Hettinga et al., 2006), the Δ blood [lactate] was not significantly different between the three pacing strategies. Although the relationship between blood [lactate] and energy yield through anaerobic glycolysis is tenuous (because the measured blood [lactate] reflects, to a greater or lesser extent, the rates of muscle lactate production, lactate efflux from muscle to blood and clearance of lactate from the blood; Brooks, 1991), the similar end-exercise blood [lactate] despite differences in time to exhaustion between the conditions might be interpreted to indicate that the *rate* of energy supply from anaerobic glycolysis (and therefore the rate of muscle and blood H^+ and lactate accumulation) was reduced in the FS condition. A reduction in the rate of H^+ accumulation might be one mechanism by which muscle fatigue is delayed and exercise tolerance is enhanced during high-intensity exercise (Bishop et al., 2002; Foster et al., 2003 cf. Westerblad & Allen, 2003; Robergs et al., 2004). The total work done above CP (i.e. W') was significantly higher during FS compared with ES and SS in the present study. Although W' is often considered to represent a fixed quantity of energy stored in O_2 , the high-energy phosphates (ATP and PCr), and a source related to anaerobic glycolysis (Monod & Scherrer, 1965; Moritani et al., 1981; Poole et al., 1988; Miura et al., 1999, 2000), it might also be related to the attainment of a critical concentration of one or more substrates or metabolites that have been linked to muscle fatigue (Coats et al., 2003; Fukuba et al., 2003; Jones et al., 2003b). The higher W' observed in FS might therefore be related not to a greater energy transfer from non-oxidative processes, but rather from a retardation of the rate of, for example, PCr depletion, or H^+ , P_i , or extra-cellular K^+ accumulation (Westerblad & Allen, 2003; Robergs et al., 2004). This effect could be a consequence of the additional O_2 consumption in FS, which effectively allowed more work to be done

above CP. It should be noted, however, that this effect is functionally indistinguishable from a pacing-strategy-induced increase in CP.

This influence of pacing strategy on O_2 consumption and exercise tolerance appears to have similarities to the effects of “priming” exercise (Jones et al., 2003b; Burnley et al., 2005). It has been shown that O_2 consumption and time to exhaustion are significantly increased during near-maximal intensity constant work-rate exercise when it is preceded by a bout of heavy-intensity exercise and a recovery period (such that blood [lactate] is mildly elevated), but whether this effect can be attributed to an increased CP, an increased W' , or both is unclear (Jones et al., 2003b; Burnley et al., 2005).

Influence of pacing strategy on AOD

Several previous studies have reported that the AOD is not altered by differences in pacing strategy (Gastin et al., 1995; Bishop et al., 2002; Foster et al., 2003, 2004), results that are seemingly consistent with the notion that there is a finite capacity for anaerobic energy supply during high-intensity (>CP) exercise. It should be cautioned, however, that the AOD concept is fraught with numerous conceptual assumptions and that its validity as a measure of anaerobic energy yield is debated (Medbo et al., 1988; Bangsbo, 1998). In the present study, the AOD was not significantly different between ES and SS in which the time to exhaustion was similar. However, during FS, the AOD was significantly *less* than was observed in the other two pacing strategies, an observation that is clearly not consistent with the similar Δ blood [lactate] measured for the three pacing strategies. We do not interpret the smaller AOD to indicate that the anaerobic energy yield was smaller for FS; indeed, the total work done above the CP was higher in FS than in the other conditions (see earlier discussion). Rather, we suggest that the smaller AOD in FS was a function of the greater exercise time in this condition, which possibly enabled the so-called “slow component” of $\dot{V}O_2$ to become more fully developed (Gaesser & Poole, 1996).

The $\dot{V}O_2$ slow component is believed to be a manifestation of a reduction in muscle efficiency and typically results in a $\dot{V}O_2$ that stabilizes at a higher value than would be predicted for the external work rate during heavy-intensity exercise (above the lactate threshold but below CP), or in a $\dot{V}O_2$ profile that continues to rise with time during constant-work-rate severe-intensity exercise (above the CP), (Poole et al., 1988; Gaesser & Poole, 1996; Özyener et al., 2001; Wilkerson et al., 2004a). The $\dot{V}O_2$ slow component becomes discernible after approximately 2 min during high-intensity exercise (Özyener et al., 2001; Wilkerson et al., 2004a,b). In the FS trial in the

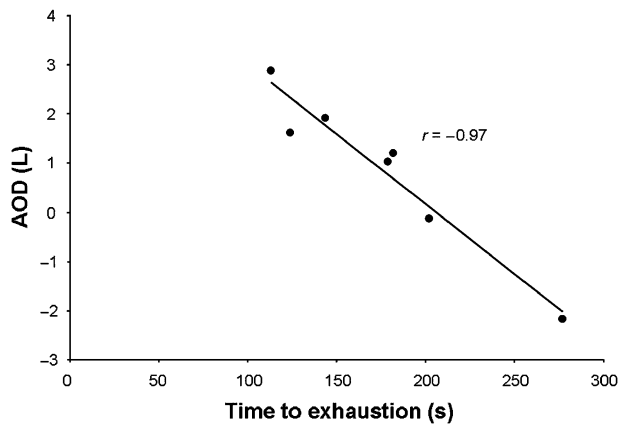


Fig. 6. Relationship between the total accumulated oxygen deficit (AOD) and the time to exhaustion in the fast-start condition. The correlation coefficient of $r = -0.97$ was statistically significant ($P < 0.001$). Note that the AOD actually became negative in the two subjects with the longest times to exhaustion.

present study, $\dot{V}O_2$ reached its highest value after approximately 90 s but then remained at this level until the end of exercise although the work rate was falling throughout the trial (Fig. 3). Similar effects have been reported previously (Stoudemire et al., 1996; Burnley et al., 2006). Therefore, the AOD that was incurred earlier in the exercise trial would have become gradually reduced as the O_2 demand (as reflected in the work rate) fell with time beyond 120 s of exercise and the O_2 uptake remained at its peak value (such that $\dot{V}O_2$ became increasingly *higher than* the expected O_2 requirement), (see lowest panel in Fig. 1 for schematic illustration of this effect). Our findings that the magnitude of the AOD in the FS condition was inversely related to the time to exhaustion and that the AOD actually became negative in two subjects (Fig. 6) confirms the earlier observations of Özyener et al. (2003), and supports their conclusion that the existence of the $\dot{V}O_2$ slow component during exercise $>CP$ confounds the calculation of AOD and challenges the validity of the AOD concept.

Limitations

The use of exponential curve fitting to describe and compare the $\dot{V}O_2$ responses to imposed work rates assumes that the ATP turnover rate is both constant over time within conditions and identical across conditions. In the present study, although the average work rate was identical for the first 120 s in the three pacing strategies, it was (by design) changing with time in both SS and FS, such that the muscle ATP turnover rate was not constant. Furthermore, even during constant-work-rate exercise (such as in the ES condition), the assumption that the ATP

turnover rate is constant from the onset of exercise has been challenged (Krustrup et al., 2003). However, although the MRT values reported in the present study should be interpreted in this light, there is no question that the total O_2 consumed was greater in the FS condition compared with the other two conditions.

Another assumption of the present study is that muscle efficiency was similar in the three conditions. However, it is presently unclear whether aerobic and anaerobic energy provision is equally “efficient” (Gladden & Welch, 1978; Krustrup et al., 2003) and so changes in the relative contribution of aerobic and anaerobic energy supply to energy turnover might have led to differences in muscle efficiency. Moreover, if muscle fiber recruitment profiles had differed appreciably between the three conditions, muscle efficiency might not have been constant either between pacing conditions or over time within a pacing condition. It has been suggested that type II fibers have both a greater ATP cost of force development and a greater phosphate cost of ATP production compared with type I fibers and that this might influence various aspects of the $\dot{V}O_2$ response to exercise (see Jones et al., 2005, for a review). A recruitment-mediated reduction in efficiency would increase the ATP and/or O_2 cost of exercise and could explain at least some of the greater O_2 consumed during the first 120 s of exercise in FS (though, importantly, *not* the enhanced exercise tolerance). Also, it is possible that differences in fiber recruitment might have enabled some fibers to utilize more (or less) of their anaerobic energy sources in the different pacing conditions.

A final consideration is the extent to which psychological factors might have influenced our results. Although our subjects were well-motivated and achieved peak $\dot{V}O_2$ and HR values that were similar to those reached during the ramp incremental test, the extent to which the different work rate profiles (i.e., descending with time in FS vs ascending with time in SS vs remaining constant with time in ES) might have altered effort perception and the distribution of the remaining physiological resources (St Clair Gibson et al., 2006) is unclear.

Applications

The results of the present study suggest that an FS pacing strategy might confer a physiological and performance advantage to athletes competing in high-intensity sport activities. It is therefore of interest that an FS pacing strategy appears to be naturally selected by athletes competing in a variety of sports including the 800 m track race (Thomas et al., 2005), sprint kayaking (Bishop et al., 2002), 2000 m rowing (Garland, 2005), 500–4000 m cycling (Foster et al.,

2003, 2004), and speed skating (van Ingen Schenau et al., 1990; Foster et al., 1994). It is unclear whether athletes select an FS pacing strategy for physiological or other (tactical, psychological, biomechanical) reasons, but the selection of a FS strategy might be considered even more remarkable given that additional power output and energy expenditure will be required to overcome inertia when exercise commences from a stationary start (de Koning et al., 1999; Garland, 2005). It might be speculated that this predisposition to “start fast” is the product of an evolutionary process – both prey and predators might gain an advantage by starting fast (and most certainly be disadvantaged by starting slow!).

The present study was designed to shed some light on the physiological mechanisms (i.e. changes in $\dot{V}O_2$ kinetics and allocation of “anaerobic” energy resources), which might underpin the improvements in performance that have been reported during high-intensity exercise when a “fast-start” or “all-out” strategy has been employed (de Koning et al., 1999; Bishop et al., 2002). However, we would caution against the temptation to directly apply our results to high-level sporting activities. Although our results appear to be promising, further research is required before firm conclusions can be drawn on the potential benefits of a fast-start strategy to the performance of high-intensity exercise. This further research should utilize protocols which more closely reflect the “real world” situation and which enable performance differences to be more rigorously compared. Moreover, it is noteworthy that the subjects in the present study did not complete warm-up exercise before any of the trials. We wished to avoid the well-known effects of such “priming” exercise on $\dot{V}O_2$ kinetics (Germino et al., 1996; Jones et al., 2003a,b; Wilkerson et al., 2004b) because this would likely have obscured our interpretation of the effects of pacing strategy, *per se*, in shaping the $\dot{V}O_2$ response dynamics. We recognize that competitive athletes would always perform prior “warm-up” exercise and it therefore remains to be determined whether, and to what extent, the effects of pacing strategy on $\dot{V}O_2$ kinetics and exercise performance persist when athletes have completed standardized warm-up activities.

There are a number of other points that should be raised with respect to the implications and applications of our study to athletic competition. Firstly, although FS resulted in a profound improvement in time to exhaustion in our study, the effect on performance in time trials or races over a fixed distance will likely be much smaller (approximately 1/16 of the effect; Hopkins et al., 1999), although probably still worthwhile. Secondly, the results of this study should not be extrapolated to lower intensity, longer duration events where the incurrence of a substantial O_2

deficit early in exercise (with consequent perturbations to muscle metabolism and blood acid–base status) might result in premature fatigue and impaired exercise performance (Demarle et al., 2001; Mattern et al., 2001). Thirdly, it should be borne in mind that the potential benefits accrued from employing an FS strategy might be offset by the additional energy requirement of overcoming air resistance if the FS results in the athlete taking the lead (Pugh, 1971). Finally, it is important that the athlete employs race tactics that optimize the distribution of the available W' relative to their CP (Fukuba & Whipp, 1996) and with consideration of the physiological profiles of the opponents (Fukuba & Whipp, 1999; Jones & Whipp, 2002).

In summary, during supra-maximal exercise bouts in which the complete utilization of the W' and therefore the time to exhaustion were predicted from the CP concept to be 120 s, a fast-start pacing strategy resulted in enhanced exercise tolerance. The fast-start strategy was associated with faster overall $\dot{V}O_2$ dynamics and a greater total O_2 consumption in the first 120 s of exercise, but the end-exercise $\dot{V}O_2$ and the Δ blood [lactate] were not significantly different between the conditions. It appears that a fast-start strategy might enhance tolerance to supra-maximal exercise by enabling a greater contribution of oxidative metabolism to total energy turnover and thereby extending the time before the finite capacity for non-oxidative ATP supply is exhausted. However, further research is required before firm conclusions can be reached on the possible benefits of a fast-start strategy to athletic performance.

Perspectives

The selection of pacing strategy during exercise has received scant attention in the scientific literature. The received wisdom is that an even-pace strategy should be optimal for performance. However, it is evident that athletes in a variety of sports (where the event duration is less than about 6 min) often choose a “fast-start” pacing strategy, and recent studies have indicated that such strategies might indeed enhance performance during high-intensity exercise. The present study helps to provide an explanation for this effect: a fast-start strategy resulted in a faster rise in $\dot{V}O_2$ following the onset of exercise such that more of the energy requirement was supplied by oxidative processes over the first 2 min of exercise. In sports events where the exercise intensity is above the CP or critical velocity, this would be expected to spare the finite anaerobic capacity and permit exercise time to be extended (or allow a higher work rate or running speed to be sustained). Athletes and coaches should consider the possible effects of pace selection along-

side other tactical considerations when attempting to optimize race performance.

Key words: $\dot{V}O_2$ kinetics, critical power, anaerobic work capacity, exercise performance, endurance, energetics.

Acknowledgements

The authors would like to thank Matthew J. Robertson and Claire L. Roberts for assistance with data collection, and two anonymous reviewers for their helpful suggestions.

References

- Aisbett B, le Rossignol P, Sparrow WA. The influence of pacing during 6-minute supra-maximal cycle ergometer performance. *J Sci Med Sport* 2003; 6: 187–198.
- Ariyoshi M, Tanaka H, Kanamori K, Obara S, Yoshitake H, Yamaji K, Shephard RJ. Influence of running pace upon performance: effects upon oxygen intake, blood lactate, and rating of perceived exertion. *Can J Appl Spt Sci* 1979a; 4: 210–213.
- Ariyoshi M, Yamaji K, Shephard RJ. Influence of running pace upon performance: effects upon treadmill endurance time and oxygen cost. *Eur J Appl Physiol Occup Physiol* 1979b; 41: 83–91.
- Bangsbo J. Quantification of anaerobic energy production during intense exercise. *Med Sci Sports Exerc* 1998; 30: 47–52.
- Bishop D, Bonetti D, Dawson B. The influence of pacing strategy on $\dot{V}O_2$ and supramaximal kayak performance. *Med Sci Sports Exerc* 2002; 34: 1041–1047.
- Brooks GA. Current concepts in lactate exchange. *Med Sci Sports Exerc* 1991; 23: 895–906.
- Burnley M, Doust JH, Jones AM. Effects of prior warm-up regime on severe-intensity cycling performance. *Med Sci Sports Exerc* 2005; 37: 838–845.
- Burnley M, Doust JH, Vanhatalo A. A 3-min all-out test to determine peak oxygen uptake and the maximal steady state. *Med Sci Sports Exerc* 2006; 38: 1995–2003.
- Coats EM, Rossiter HB, Day JR, Miura A, Fukuba Y, Whipp BJ. Intensity-dependent tolerance to exercise after attaining $\dot{V}O_2$ max in humans. *J Appl Physiol* 2003; 95: 483–490.
- de Koning JJ, Bobbert MF, Foster C. Determination of optimal pacing strategy in track cycling with an energy flow model. *J Sci Med Sport* 1999; 2: 266–277.
- Demarle AP, Slawinski JJ, Laffite LP, Bocquet VG, Koralsztein JP, Billat VL. Decrease of O_2 deficit is a potential factor in increased time to exhaustion after specific endurance training. *J Appl Physiol* 2001; 90: 947–953.
- Foster C, de Koning JJ, Hettinga F, Lampen J, Dodge C, Bobbert M, Porcari JP. Effect of competitive distance on energy expenditure during simulated competition. *Int J Sports Med* 2004; 25: 198–204.
- Foster C, de Koning JJ, Hettinga F, Lampen J, La Clair KL, Dodge C, Bobbert M, Porcari JP. Pattern of energy expenditure during simulated competition. *Med Sci Sports Exerc* 2003; 35: 826–831.
- Foster C, Schrager M, Snyder AC, Thompson NN. Pacing strategy and athletic performance. *Sports Med* 1994; 17: 77–85.
- Foster C, Snyder AC, Thompson NN, Green MA, Foley M, Schrager M. Effect of pacing strategy on cycle time trial performance. *Med Sci Sports Exerc* 1993; 25: 383–388.
- Fukuba Y, Miura A, Endo M, Kan A, Yanagawa K, Whipp BJ. The curvature constant parameter of the power-duration curve for varied-power exercise. *Med Sci Sports Exerc* 2003; 35: 1413–1418.
- Fukuba Y, Whipp BJ. The “endurance parameter ratio” of the power-duration curve and race variation strategy for distance running. In: Steinacker P, Ward SA, eds. *The physiology and pathophysiology of exercise tolerance*. New York: Plenum Press, 1996: 321–328.
- Fukuba Y, Whipp BJ. A metabolic limit on the ability to make up for lost time in endurance events. *J Appl Physiol* 1999; 87: 853–861.
- Gaesser GA, Poole DC. The slow component of oxygen uptake kinetics in humans. *Exerc Sport Sci Rev* 1996; 24: 35–71.
- Garland SW. An analysis of the pacing strategy adopted by elite competitors in 2000 m rowing. *Br J Sports Med* 2005; 39: 39–42.
- Gastin PB, Costill DL, Lawson DL, Krzeminski K, McConell GK. Accumulated oxygen deficit during supramaximal all-out and constant intensity exercise. *Med Sci Sports Exerc* 1995; 27: 255–263.
- Gerbino A, Ward SA, Whipp BJ. Effects of prior exercise on pulmonary gas-exchange kinetics during high-intensity exercise in humans. *J Appl Physiol* 1996; 80: 99–107.
- Gladden LB, Welch HG. Efficiency of anaerobic work. *J Appl Physiol* 1978; 44: 564–570.
- Hettinga FJ, de Koning JJ, Broersen FT, Van Geffen P, Foster C. Pacing strategy and the occurrence of fatigue in 4000-m cycling time trials. *Med Sci Sports Exerc* 2006; 38: 1484–1491.
- Hopkins WG, Hawley JA, Burke LM. Design and analysis of research on sport performance enhancement. *Med Sci Sports Exerc* 1999; 31: 472–485.
- Hughson RL, O’Leary DD, Betik AC, Hebestreit H. Kinetics of oxygen uptake at the onset of exercise near or above peak oxygen uptake. *J Appl Physiol* 2000; 88: 1812–1819.
- Jones AM, Koppo K, Burnley M. Effects of prior exercise on metabolic and gas exchange responses to exercise. *Sports Med* 2003a; 33: 949–971.
- Jones AM, Pringle JSM, Carter H. Influence of muscle fibre type and motor unit recruitment on $\dot{V}O_2$ kinetics. In: Jones AM, Poole DC, eds. *Oxygen uptake kinetics in sport, exercise and medicine*. London: Routledge, 2005: 261–293.
- Jones AM, Whipp BJ. Bioenergetic constraints on tactical decision making in middle distance running. *Br J Sports Med* 2002; 36: 102–104.
- Jones AM, Wilkerson DP, Burnley M, Koppo K. Prior heavy exercise enhances performance during subsequent perimaximal exercise. *Med Sci Sports Exerc* 2003b; 35: 2085–2092.
- Krustrup P, Ferguson RA, Kjaer M, Bangsbo J. ATP and heat production in human skeletal muscle during dynamic exercise: higher efficiency of anaerobic than aerobic ATP resynthesis. *J Physiol* 2003; 549: 255–269.
- Leger LA, Ferguson RJ. Effect of pacing on oxygen uptake and peak lactate for a mile run. *Eur J Appl Physiol Occup Physiol* 1974; 32: 251–257.
- Mahler M. First-order kinetics of muscle oxygen consumption, and an equivalent proportionality between QO_2 and phosphorylcreatine level.

- Implications for the control of respiration. *J Gen Physiol* 1985: 86: 135–165.
- Margaria R, Mangili F, Cuttica F, Cerretelli P. The kinetics of the oxygen consumption at the onset of muscular exercise in man. *Ergonomics* 1965: 8: 49–54.
- Mattern CO, Kenefick RW, Kertzer R, Quinn TJ. Impact of starting strategy on cycling performance. *Int J Sports Med* 2001: 22: 350–355.
- Medbo JJ, Mohn AC, Tabata I, Bahr R, Vaage O, Sejersted OM. Anaerobic capacity determined by maximal accumulated O₂ deficit. *J Appl Physiol* 1988: 64: 50–60.
- Meyer RA, Sweeney HL, Kushmerick MJ. A simple analysis of the “phosphocreatine shuttle”. *Am J Physiol* 1984: 246: C365–C377.
- Miura A, Kino F, Kajitani S, Sato H, Sato H, Fukuba Y. The effect of oral creatine supplementation on the curvature constant parameter of the power-duration curve for cycle ergometry in humans. *Jpn J Physiol* 1999: 49: 169–174.
- Miura A, Sato H, Sato H, Whipp BJ, Fukuba Y. The effect of glycogen depletion on the curvature constant parameter of the power-duration curve for cycle ergometry. *Ergonomics* 2000: 43: 133–141.
- Miyamoto Y, Niizeki K. Dynamics of ventilation, circulation, and gas exchange to incremental and decremental ramp exercise. *J Appl Physiol* 1992: 72: 2244–2254.
- Monod H, Scherrer J. The work capacity of a synergic muscle group. *Ergonomics* 1965: 8: 329–338.
- Moritani T, Nagata A, deVries HA, Muro M. Critical power as a measure of physical work capacity and anaerobic threshold. *Ergonomics* 1981: 24: 339–350.
- Niizeki K, Takahashi T, Miyamoto Y. A model analysis of asymmetrical response of pulmonary $\dot{V}O_2$ during incremental and decremental ramp exercise. *J Appl Physiol* 1995: 79: 1816–1827.
- Özyener F, Rossiter HB, Ward SA, Whipp BJ. Influence of exercise intensity on the on- and off-transient kinetics of pulmonary oxygen uptake in humans. *J Physiol* 2001: 533: 891–902.
- Özyener F, Rossiter HB, Ward SA, Whipp BJ. Negative accumulated oxygen deficit during heavy and very heavy intensity cycle ergometry in humans. *Eur J Appl Physiol* 2003: 90: 185–190.
- Poole DC, Ward SA, Gardner GW, Whipp BJ. Metabolic and respiratory profile of the upper limit for prolonged exercise in man. *Ergonomics* 1988: 31: 1265–1279.
- Pugh LG. The influence of wind resistance in running and walking and the mechanical efficiency of work against horizontal or vertical forces. *J Physiol* 1971: 213: 255–276.
- Robergs RA, Ghiasvand F, Parker D. Biochemistry of exercise-induced metabolic acidosis. *Am J Physiol Regul Integr Comp Physiol* 2004: 287: R502–R516.
- Robinson S, Robinson DL, Mountjoy RJ, Bullard RW. Influence of fatigue on the efficiency of men during exhausting runs. *J Appl Physiol* 1958: 12: 197–201.
- Rossiter HB, Howe FA, Ward SA. Intramuscular phosphate and pulmonary $\dot{V}O_2$ kinetics during exercise. In: Jones AM, Poole DC, eds. *Oxygen uptake kinetics in sport, exercise and medicine*. London: Routledge, 2005: 154–184.
- Scheuermann BW, Barstow TJ. O₂ uptake kinetics during exercise at peak O₂ uptake. *J Appl Physiol* 2003: 95: 2014–2022.
- St Clair Gibson A, Lambert EV, Rauch LH, Tucker R, Baden DA, Foster C, Noakes TD. The role of information processing between the brain and peripheral physiological systems in pacing and perception of effort. *Sports Med* 2006: 36: 705–722.
- Stoudemire NM, Wideman L, Pass KA, McGinnes CL, Gaesser GA, Weltman A. The validity of regulating blood lactate concentration during running by ratings of perceived exertion. *Med Sci Sports Exerc* 1996: 28: 490–495.
- Thomas C, Hanon C, Perrey S, Le Chevalier JM, Couturier A, Vandewalle H. Oxygen uptake response to an 800-m running race. *Int J Sports Med* 2005: 26: 268–273.
- Thompson KG, MacLaren DP, Lees A, Atkinson G. The effect of even, positive and negative pacing on metabolic, kinematic and temporal variables during breaststroke swimming. *Eur J Appl Physiol* 2003: 88: 438–443.
- van Ingen Schenau GJ, de Koning JJ, de Groot G. A simulation of speed skating performances based on a power equation. *Med Sci Sports Exerc* 1990: 22: 718–728.
- Westerblad H, Allen DG. Cellular mechanisms of skeletal muscle fatigue. *Adv Exp Med Biol* 2003: 538: 563–570.
- Whipp BJ, Davis JA, Torres F, Wasserman K. A test to determine parameters of aerobic function during exercise. *J Appl Physiol* 1981: 50: 217–221.
- Whipp BJ, Mahler M. Dynamics of gas exchange during exercise. In: West JB, ed. *Pulmonary gas exchange*. New York: Academic Press, 1980: Vol. II, 33–96.
- Whipp BJ, Rossiter HB. The kinetics of oxygen uptake: physiological inferences from the parameters. In: Jones AM, Poole DC, eds. *Oxygen uptake kinetics in sport, exercise and medicine*. London: Routledge, 2005: 62–94.
- Wilkerson DP, Jones AM. Influence of initial metabolic rate on pulmonary O₂ uptake on-kinetics during severe intensity exercise. *Respir Physiol Neurobiol* 2006: 152: 204–219.
- Wilkerson DP, Koppo K, Barstow TJ, Jones AM. Effect of work rate on the functional ‘gain’ of Phase II pulmonary O₂ uptake response to exercise. *Respir Physiol Neurobiol* 2004a: 142: 211–223.
- Wilkerson DP, Koppo K, Barstow TJ, Jones AM. Effect of prior multiple-sprint exercise on pulmonary O₂ uptake kinetics following the onset of perimaximal exercise. *J Appl Physiol* 2004b: 97: 1227–1236.